

Study of Prompt Engineering in In-Context Learning: Comparison of Zero-Shot, One-Shot, and Few-Shot Approaches on Text-to-Text and Image-to-Image Models

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Abstract

This study investigates the effectiveness of prompt engineering and in-context learning for large language models (LLMs) and image generation models. We systematically compare zero-shot, one-shot, and few-shot learning approaches, demonstrating that well-designed prompts significantly enhance model performance. Our results reveal that few-shot learning, combined with advanced techniques like batch processing and sliding windows, achieves the highest accuracy and CLIP scores. However, providing reference examples does not always guarantee better results, as poorly chosen examples can mislead the model. This research highlights the importance of thoughtful prompt design and the strategic use of in-context learning to improve the reliability and efficiency of AI models across various applications.

1 Introduction

In-context learning has emerged as a powerful technique for enhancing the performance of large language models (LLMs) and image generation models without requiring extensive retraining. By leveraging carefully designed prompts, this approach enables models to better understand and execute tasks, reducing instances of hallucinations, where models generate incorrect or fabricated information and improving the overall quality of outputs [1].

In this study, we explore the role of prompt engineering in improving model accuracy across both text-to-text and image-to-image tasks. We systematically compare three primary approaches: zero-shot, one-shot, and few-shot learning. While few-shot learning often yields the highest accuracy and semantic alignment [2], our findings reveal that zero-shot learning can also perform remarkably well, particularly for models with strong general knowledge. However, we also highlight that providing reference examples does not always guarantee better results; in some cases, poorly chosen examples can mislead the model, leading to suboptimal performance [3].

2 Methodology

In this paper, we systematically evaluated the performance of large language models (LLMs) and image generation models using in-context learning techniques. Our methodology focused on comparing three primary approaches: zero-shot, one-shot, and few-shot learning, across both text-to-text and image-to-image tasks.

For text-to-text tasks, we assessed the models' ability to generate accurate and contextually relevant responses. The tasks included sentiment analysis using the McDonald's review dataset and factual question answering. We evaluated the models' performance using accuracy as the primary metric.

For image generation tasks, we employed Stable

Diffusion to generate images based on text prompts. The approaches included zero-shot (no reference images), one-shot (single reference image), and few-shot (multiple reference images). We used the CLIP score to measure the semantic alignment between generated images and text prompts [4]. Advanced techniques such as sliding windows and patch processing were also explored to optimize performance.

By following this methodology, we systematically compared the effectiveness of different prompting approaches and provided insights into the strengths and limitations of in-context learning for both text and image generation tasks.

3 Text-to-Text Models

In this section, we present the results of our experiments on text-to-text tasks using GPT-3.5 Turbo, GPT-4, and Llama-3.1-8b [5]. We evaluated the performance of these models under different prompting strategies, including zero-shot, one-shot, and few-shot learning, specifically focusing on sentiment analysis using the McDonald’s review dataset. Our goal was to assess how well these models can classify sentiments accurately based on the provided prompts [6].

3.1 Experiments with GPT-3.5 Turbo

We first evaluated the performance of GPT-3.5 Turbo on sentiment analysis tasks using the McDonald’s review dataset. The results, summarized in Table 1, show that:

#	Mode	Time (s)	Accuracy (%)
0	Zero-shot	211.92	70.0
1	One-shot	177.73	75.2
2	Few-shot	183.28	76.6

Table 1: Performance comparison between zero-shot, one-shot, and few-shot modes using GPT-3.5 Turbo on the McDonald’s review dataset.

- Zero-shot learning achieved an accuracy of 70.0%, demonstrating the model’s ability to classify sentiments without any examples.

- One-shot learning improved accuracy to 75.2%, as the single example provided additional context for sentiment classification.

- Few-shot learning yielded the highest accuracy of 76.6%, showing that multiple examples further enhance the model’s ability to classify sentiments accurately.

3.2 Experiments with GPT-4

Next, we evaluated GPT-4 on the same sentiment analysis task using the McDonald’s review dataset. The results, summarized in Table 2, reveal interesting insights:

#	Mode	Time (s)	Accuracy (%)
0	Zero-shot	277.02	77.4
1	One-shot	277.24	75.6
2	Few-shot	355.15	77.6

Table 2: Performance comparison of different prompting modes using GPT-4 on the McDonald’s review dataset.

- Zero-shot learning achieved the highest accuracy (77.4%) among the three approaches, outperforming both one-shot (75.6%) and few-shot (77.6%). This is due to GPT-4’s extensive general knowledge, which allows it to classify sentiments effectively even without examples.

- Few-shot learning showed only a slight improvement over zero-shot (77.6% vs. 77.4%), indicating that additional examples provide limited benefits for GPT-4 in this context.

- One-shot learning had the lowest accuracy, suggesting that a single example may not always provide sufficient guidance and can sometimes mislead the model.

3.3 Experiments with Llama-3.1-8b

We also evaluated the Llama-3.1-8b model on the same sentiment analysis task, comparing its performance in standard processing and batch processing modes. The results are summarized in Table 3 and Table 4.

#	Mode	Time (s)	Accuracy (%)
0	Zero-shot	1246.21	75.8
1	One-shot	1195.77	72.8
2	Few-shot	1211.6	75

Table 3: Performance comparison of different prompting modes using Llama-3.1-8b in standard processing mode.

#	Mode	Time (s)	Accuracy (%)
0	Zero-shot	180.5	80
1	One-shot	242.31	79
2	Few-shot	242.6	77

Table 4: Performance comparison of different prompting modes using Llama-3.1-8b in batch processing mode.

- Standard Processing:
 - Zero-shot learning achieved an accuracy of 75.8%, demonstrating the model’s ability to classify sentiments without examples.
 - One-shot learning had a lower accuracy of 72.8%, indicating that a single example may not always improve performance.
 - Few-shot learning yielded an accuracy of 75%, showing that multiple examples can help but with limited gains.
- Batch Processing:
 - Zero-shot learning achieved the highest accuracy (80%), outperforming both one-shot (79%) and few-shot (77%). This highlights the efficiency of batch processing in improving model performance.
 - Few-shot learning showed a slight decrease in accuracy compared to zero-shot, suggesting that additional examples may not always be beneficial in batch mode.
 - One-shot learning performed well, with an accuracy of 79%, indicating that batch processing can enhance the model’s ability to generalize even with limited examples.

3.4 Comparison Between GPT-3.5 Turbo, GPT-4, and Llama-3.1-8b

When comparing the three models on the McDonald’s review dataset, we observed the following:

- GPT-4 consistently outperformed both GPT-3.5 Turbo and Llama-3.1-8b across all prompting approaches. For example, in zero-shot learning, GPT-4 achieved an accuracy of 77.4%, compared to 70.0% for GPT-3.5 Turbo and 75.8% for Llama-3.1-8b in standard processing.
- Llama-3.1-8b in batch processing mode achieved the highest zero-shot accuracy (80%), surpassing both GPT-3.5 Turbo and GPT-4. This demonstrates the effectiveness of batch processing in optimizing model performance.
- The performance gap between zero-shot and few-shot learning was smaller for GPT-4 and Llama-3.1-8b in batch mode, highlighting their superior ability to generalize without examples.

3.5 Key Findings

Our experiments with text-to-text models on the McDonald’s review dataset highlight the following key findings:

- Few-shot learning generally yields the highest accuracy, as multiple examples provide the model with better context for sentiment classification. This is particularly true for GPT-3.5 Turbo.
- Zero-shot learning can perform exceptionally well, especially for models with strong general knowledge like GPT-4 and Llama-3.1-8b in batch processing mode.
- One-shot learning provides a balance between zero-shot and few-shot approaches but can sometimes lead to ambiguous or incorrect responses if the example is not well-chosen.
- Batch processing significantly improves both accuracy and processing speed, as demonstrated by the performance of Llama-3.1-8b.

These results underscore the importance of prompt design, in-context learning, and hardware optimization in enhancing the performance of text-to-text

models, particularly in sentiment analysis tasks. They also demonstrate the advanced capabilities of GPT-4 and Llama-3.1-8b, which leverage their extensive pre-training and efficient processing to achieve high accuracy even without examples.

4 Image-to-Image Models

In this section, we present the results of our experiments on image-to-image tasks using Stable Diffusion [7]. We evaluated the model’s ability to generate high-quality images based on text prompts, comparing three primary approaches: zero-shot, one-shot, and few-shot learning. Our goal was to assess how well the model can transform textual descriptions into visually coherent and semantically aligned images [8].

4.1 Zero-Shot Learning

In the zero-shot learning approach, the model generates images solely based on a text prompt, without any reference images. This method tests the model’s ability to interpret and visualize textual descriptions from scratch.

- **Observations:** The model demonstrated an impressive ability to generate visually appealing images that align well with the provided text prompts. However, certain complex details or nuanced scenes were less precise, highlighting the limitations of zero-shot learning in capturing intricate elements without additional guidance. Zero-shot learning is particularly useful for generating creative and diverse outputs, but it may struggle with highly specific or detailed prompts.

4.2 One-Shot Learning

In the one-shot learning approach, the model is provided with a single reference image alongside the text prompt. This reference image serves as a guide, helping the model better understand the context and style required for the output.

- **Observations:** The inclusion of a reference image significantly improved the model’s ability to generate more detailed and contextually accurate im-

ages. However, the reference image can also **divert the model** from the intended output. If the reference image contains elements that are not fully aligned with the text prompt, the generated image may incorporate these irrelevant details, leading to a divergence from the desired result. The quality of the output heavily depended on the relevance and quality of the reference image. A well-chosen reference image led to more coherent and visually aligned results, while an unsuitable reference could degrade the output.

4.3 Few-Shot Learning

In the few-shot learning approach, the model is provided with multiple reference images alongside the text prompt. This method leverages a richer context, allowing the model to combine elements from multiple examples to generate the final output. To optimize this process, we employed techniques such as **sliding windows** and **patch processing**, which helped the model efficiently handle and integrate multiple references.

- **Observations:** Few-shot learning can produce highly detailed and visually coherent images, as the multiple reference images provide a broader and more nuanced context for the model to draw from. However, **adding more reference images does not always lead to better results**. In some cases, the model may struggle to reconcile conflicting or inconsistent elements from the reference images, leading to outputs that diverge from the intended goal. For example, if the reference images introduce contradictory styles or details, the generated image may become less coherent or fail to align with the text prompt.

4.4 Evaluating Image Quality with CLIP Scores

To quantify the quality of the generated images, we used CLIP scores, which measure the semantic alignment between the generated images and their corresponding text prompts [9]. We evaluated three approaches: zero-shot, one-shot, and few-shot learning alongside two specific few-shot methods: Method 2

(Resize) and Method 3 (Sliding Window). The results are summarized in Table 5.

Approach	CLIP Score
Zero-Shot	0.3687
One-Shot	0.3445
Few-Shot (Resize)	0.3589
Few-Shot (Sliding Window)	0.4221

Table 5: CLIP scores for zero-shot, one-shot, and few-shot approaches (including Resize and Sliding Window methods).

- **Zero-Shot:** Achieved a CLIP score of 0.3687, demonstrating the model’s ability to generate images that align with the text prompts without examples. This approach is flexible but may struggle with complex or highly specific prompts.

- **One-Shot:** Achieved a slightly lower CLIP score of 0.3445, indicating that a single reference image may not always improve semantic alignment. The quality of the output depends heavily on the relevance of the reference image.

- **Few-Shot (Resize):** Achieved a CLIP score of 0.3589. In this method, reference images are resized to fit the model’s input requirements. However, resizing causes a significant loss of information, particularly in fine-grained details. This loss of information limits the model’s ability to fully utilize the reference images, resulting in a moderate improvement over one-shot learning but falling short of the potential offered by more advanced techniques.

- **Few-Shot (Sliding Window):** Achieved the highest CLIP score of 0.4221, significantly outperforming the other methods. This method uses a sliding window approach to process multiple reference images. Here’s how it works: - Process: Each reference image is divided into smaller patches using a sliding window technique. The model generates an image for each patch, and the best result (highest CLIP score) is selected as the final output. - Why it works: The sliding window approach significantly improves few-shot performance because it allows the model to focus on the most relevant parts of the reference images. By processing the images in patches, the model can better capture fine-grained details and

maintain semantic alignment with the text prompt. This method also benefits zero-shot and one-shot approaches, but the improvement is most noticeable in few-shot learning, where multiple references are effectively utilized.

5 Conclusion

In this study, we explored the effectiveness of **prompt engineering** and **in-context learning** across both **text-to-text** and **image-to-image** tasks. By systematically comparing **zero-shot**, **one-shot**, and **few-shot** learning approaches, we demonstrated that well-designed prompts can significantly enhance model performance. However, our results also revealed that providing reference examples does not always guarantee better results, as poorly chosen examples can mislead the model rather than improve its accuracy. This is particularly true for generative models, which already possess a broad general knowledge base.

In conclusion, our research underscores the importance of **thoughtful prompt design** and **strategic use of in-context learning** in enhancing the reliability and efficiency of AI models. While few-shot learning with advanced techniques like sliding windows offers significant improvements, zero-shot learning remains a powerful and flexible approach, particularly for models with strong general knowledge. These findings have practical implications for industries relying on AI, such as content creation, customer service, and healthcare, where accurate and efficient model performance is critical.

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